■ EASYWAY

Unified Urban Mobility & Food Delivery Super-App (EasyWay Rides & EasyWay Food)

Complete Technical, Functional & Architectural Specification

An end-to-end full-stack platform combining real-time symmetrical ride dispatch, AI voice assistant booking, 3-route navigation, and a 5-stage live kitchen preparation pipeline.

Document Version: 1.0.0 (Production Blueprint)
Stack: Node.js, Express, TypeScript (ESM), Socket.io, PostGIS / In-Memory Spatial, Leaflet.js
Author / Organization: EasyWay Engineering Team
Status: Live & Operational (Local Port: 4000)
Target Geography: Vijayawada Urban Hub (Extendable Globally)

■ **Core Architectural Highlights**

- **Symmetrical Canonical Fares:** Guaranteed Fare(A → B) ≡ Fare(B → A) with zero price asymmetry.
- **AI Voice Assistant:** Speech recognition, NLP landmark parsing & 2-way text-to-speech audio feedback.
- **3-Way Alternative Routing:** Interactive Way 1 (Fastest), Way 2 (Eco), and Way 3 (Scenic) navigation.
- **Dual-Mode Persistence:** Native PostGIS geospatial indexing + In-Memory Haversine fallback.
- **5-Stage Kitchen Pipeline:** Real-time multi-vendor food preparation and OTP delivery tracking.

1. System Architecture & Technology Stack

EasyWay is designed as a unified, single-binary Super-App. The platform integrates two decoupled business domains (EasyWay Rides & EasyWay Food) through a high-concurrency event bus and spatial engine.

Layer	Technology Components	Key Role / Function
Backend Core	Node.js 18+, Express.js 4.19, TypeScript 5.5	REST APIs, middleware, type safety & orchestration
Real-Time	Socket.io 4.7, WebSockets	Driver dispatch ring, live GPS streaming, kitchen sync
Spatial Engine	PostGIS / In-Memory Haversine Math	Spherical distance calculation, geocoding & driver proximity
Voice Engine	Web Speech API & NLP Parser	Speech recognition, entity extraction & 2-way TTS audio
Frontend UI	Leaflet.js 1.9.4, Modern Glassmorphism CSS	Interactive maps, split-screen live demos, cart drawer
Security	JWT (jsonwebtoken), Zod validation	Token auth, role-based access & payload validation

2. Platform 1: Rides & Mobility (EasyWay Rides)

2.1. AI Voice Booking Assistant

The AI Voice Booking Assistant provides hands-free trip dispatch via browser speech recognition and heuristic entity extraction. It supports voice-driven origin/destination selection, vehicle switching, and route inversions.

- **Visual Status Orb:** Displays animated states for Idle (Blue), Listening (Red pulse), Parsing (Amber spin), Speaking (Green wave), and Mic Blocked (Orange).
- **2-Way Spoken Speech Output (TTS):** Speaks trip confirmations, estimated arrival times, and locked fares in clear natural speech.
- **1-Click Quick Voice Chips:** Fallback interactive chips allowing instant execution even in environments without microphone permissions.
- **Voice Features & Guide Modal:** Built-in documentation modal with a live command runner.

2.2. Symmetrical Canonical Fare Guarantee

Traditional ride platforms apply asymmetric pricing where traveling from Point A to Point B costs differently than B to A. EasyWay eliminates price discrimination by enforcing exact symmetry: **Fare(A → B) ≡ Fare(B → A)**.

Vehicle Category	Base Fare	Per Km Rate	Per Min Rate	Platform Fee
■ Bike Taxi (2-Wheeler)	■15.00	■6.50	■1.00	■3.00
■ Auto Rickshaw (3-Wheeler)	■25.00	■9.00	■1.25	■5.00
■ Prime Cab (4-Wheeler)	■50.00	■14.00	■1.75	■10.00

2.3. Safety, Security & Invoice Generation

- **4-Digit Start OTP:** Trips cannot begin until the Captain enters the rider's confidential OTP into their driver terminal.
- **Anti-Cancellation Guarantee:** Once accepted, driver cancellation is locked out to prevent passenger stranding.
- **Itemized Tax Invoices:** Generates official GST-compliant tax invoices with PDF download and email dispatch actions.
- **Emergency SOS Safety Center:** Loud browser audio siren, live encrypted GPS link sharing, and instant police emergency dispatch.

3. Platform 2: Food Delivery (EasyWay Food)

3.1. Restaurant Catalog & Cart Drawer

EasyWay Food features an integrated multi-vendor marketplace with 13 pre-seeded restaurants (including Paradise Biryani, Minerva Grand, Bawarchi, Domino's, and Haldiram's). It includes a slide-out cart drawer with itemized billing (packaging fee, delivery fee, platform fee, and 5% GST).

3.2. 5-Stage Real-Time Kitchen Preparation Pipeline

Every order transitions through a state machine synchronized across the Kitchen Manager, Delivery Partner, and Customer view in real time:

Stage	State Name	Operational Action / Trigger
1	**PLACED**	Customer completes online prepaid checkout; order enters kitchen queue.
2	**IN_KITCHEN**	Head chef accepts order; food preparation & cooking starts.
3	**READY_FOR_PICKUP**	Cooking completed; food boxed with thermal tamper seals.
4	**OUT_FOR_DELIVERY**	Delivery partner picks up parcel from restaurant after verification.
5	**DELIVERED**	Delivery partner inputs customer's 4-digit Handover OTP at doorstep.

4. REST API & WebSocket Specifications

Method	Endpoint	Description
GET	/api/v1/health	Returns service health, uptime & PostGIS status.
POST	/api/v1/fare/calculate	Calculates deterministic symmetrical fare for pickup & drop.
POST	/api/v1/rides/voice-book	AI Voice pipeline: NLP geocoding, route calc & auto-dispatch.
POST	/api/v1/rides/request	Creates ride request and broadcasts offer to nearby captains.
POST	/api/v1/rides/:id/accept	Captain accepts ride offer; triggers anti-cancellation lock.
POST	/api/v1/rides/:id/verify-otp	Verifies 4-digit ride start OTP and begins trip telemetry.
POST	/api/v1/rides/:id/complete	Completes trip; calculates final bill & triggers tax invoice.
GET	/api/v1/food/restaurants	Fetches all 13 restaurant profiles, dishes & ratings.
POST	/api/v1/food/orders	Creates prepaid food order & adds to live kitchen queue.

5. Setup, Verification & Test Credentials

Quick Start Commands:

```
cd backend
npm install
npm start
# App runs on: http://localhost:4000
# Test Console: http://localhost:4000/test-console
```

Pre-Seeded Test Credentials (Dev OTP: 123456):

Role / User	Phone Number	Default Vehicle / Capabilities
Customer / Rider (Rahul)	9876543210	Rides booking, Voice Orb, Food cart & tracking
Bike Captain (Ramesh)	9876543211	Two-Wheeler Bike Taxi dispatch terminal
Auto Captain (Suresh)	9876543212	Three-Wheeler Auto Rickshaw dispatch terminal
Cab Captain (Vikram)	9876543213	Four-Wheeler Prime Cab dispatch terminal